Causal Token Pruning for Streaming Video VLMs

M1 + TAST: a training-free, cross-model stack at 30% LLM-token compute

Srisashank Potluru
George Mason University
spotluru@gmu.edu

2026-04-19

Abstract

We address training-free token pruning for streaming video VLMs. Starting from StreamingVLM, we identify five structural failure modes that cap naive multi-layer ViT salience pruning at the vanilla accuracy ceiling (82.08% on DeViBench-360p), and propose a two-module stack — DyMU score-weighted nearest-kept merge (M1) plus Temporal Accumulative State Tokens (TAST) — that runs strictly causally inside the existing online `inference.py` loop. On StreamingVLM, our stack matches vanilla accuracy at $\approx$ 30% of LLM-token compute (82.21% with 95% CI $[79.14, 85.12]$, n=652), where the prior 11%-budget setting drops to 80.21%. Crucially, the same stack — *without retraining* — generalizes across the model axis to Qwen2.5-VL-7B-Instruct, where it reaches **86.50%** on the same DeViBench-360p split (95% CI $[83.90, 88.96]$, +4.42 pp over the StreamingVLM vanilla anchor with sub-vanilla compute). On MVBench, a paired-bootstrap comparison on the same 1325-sample subset yields +3.55 pp (95% CI $[+1.43, +5.58]$) over vanilla StreamingVLM at $\sim$ 30% LLM-token compute — the only short-video delta in our sweep that survives 95% bootstrap CI. On the long-video MLVU plotQA split we find a statistically significant -7.00 pp loss (95% CI $[-10.00, -4.20]$), isolating long-horizon temporal aggregation as the key open problem. We further evaluate stochastic top-k self-consistency at the same compute budget: a 3-seed Gumbel-noise vote (σ=0.05) recovers 82.36% and identifies sample diversity at $\sigma \leq 0.05$ as the primary limiter. We release the full 14-variant ablation, bootstrap CIs, and three benchmark-eval scripts patched to subsample reproducibly.

1 Introduction

Streaming video understanding requires a vision-language model (VLM) to consume an unbounded sequence of frames and answer questions in real time, without re-processing past content. The dominant cost in production VLMs is the language-model decoder, which scales with the total visual token count fed in from the vision encoder. For a Qwen2.5-VL-7B-class model on a 30 fps, 360p stream this routinely exceeds 10k tokens per minute — intractable for online deployment.

Token pruning works on images; streaming video is harder. Image-domain methods such as FastV, CDPruner, and DyMU achieve 10×–20× token reductions while preserving accuracy, by exploiting spatial redundancy. Naively porting these to video collapses for two reasons: (i) per-frame independent pruning destroys the temporal evidence the VLM needs for action and counting questions, and (ii) global redundancy penalties (MMR, DPP) treat the same object viewed across frames as duplicate, killing the signal exactly when it matters.

Our contributions.

1. We expose five structural failure modes of multi-layer ViT salience pruning on streaming video — including a silent CLI-flag override that invalidated all our earlier ablations — and document a 14-variant negative sweep (Section 3.5).

2. We propose a minimal two-module stack: **M1** (DyMU score-weighted nearest-kept merge) and **TAST** (Temporal Accumulative State Tokens), which compose with strict causality and zero retraining (Section 3).

3. On MVBench we measure a paired-bootstrap delta of **+3.55 pp** (95% CI $[+1.43, +5.58]$) over vanilla StreamingVLM at $\sim 30\%$ of LLM-token compute, the only statistically certified short-video win in our sweep.

4. Cross-model: *the same training-free stack transfers to* Qwen2.5-VL-7B-Instruct, where it reaches 86.50% on DeViBench-360p (+4.42 pp over the streaming baseline anchor, lower-CI 83.90%).

5. Honest negative result: on long-form video (MLVU plotQA, 15–30 min) the same stack loses -7.00 pp (95% CI $[-10.00, -4.20]$), isolating long-horizon temporal aggregation as the next open problem.

2 Related Work

Visual token pruning for VLMs. FastV [?] drops half the visual tokens after the second LLM layer based on attention scores. CDPruner [?] reformulates pruning as conditional DPP MAP and reports retaining 94% of vanilla accuracy at 11.1% of the token budget on image benchmarks. DyMU [?] merges dropped tokens into nearest-kept neighbours rather than discarding them; we adapt this idea (M1) to the multi-layer ViT-salience setting and observe it is necessary but not sufficient for video.

Streaming and long-form video VLMs. StreamingVLM [?] extends Qwen2.5-VL-7B with chunked attention and a fixed-size visual KV cache for online streams. MovieChat and LongVA take similar long-context approaches by training new position embeddings or memory tokens. Our TAST module is closest in spirit to LongVA's memory tokens but is training-free, with n=32 EMA-updated slots reusable across any base VLM.

Temporal salience and self-consistency. STAR [?] and STAMP [?] score per-token salience from the vision encoder. Self-consistency for VLMs has been studied for multiple-choice QA [?]; we adapt it under a strict *compute-equal* constraint by running k=3 stochastic top-k prunings at r=0.30, keeping total LLM tokens at 0.90× vanilla.

Position of this work. Prior streaming VLM work focuses on *accuracy at vanilla compute*; prior pruning work focuses on *retention at aggressive compute on images*. We sit at the intersection: training-free, causal, $\sim 30\%$ LLM-token budget on streaming video, with statistically certified deltas on a benchmark (MVBench) where retention-style claims have not previously been tested.

3 Method

We present three orthogonal modules that compose into a streaming-friendly training-free pruning stack: (i) **STAMP-Temporal**, a multi-layer ViT salience filter that keeps a fraction r of tokens per chunk; (ii) **M1 / DyMU-merge**, a score-weighted nearest-kept merge that recovers information from

dropped tokens at zero extra LLM cost; and (iii) **TAST** (Temporal Accumulative State Tokens), a small EMA-maintained pool of n=32 summary tokens injected after the kept-token sequence to carry motion evidence across chunk boundaries. All three operate strictly causally — no future-frame access — and run unchanged in the online `inference.py` setup.

3.1 STAMP-Temporal

For each incoming chunk c of T frames, we extract per-token salience from ViT layers $\mathcal{L}=\{7, 15, 23, 31\}$ and average them with EMA decay α=0.5 across chunks, $S_t=\alpha\,\bar{A}_t + (1-\alpha)\,S_{t-1}$, where $\bar{A}_t$ is the per-token mean attention from `[CLS]` across $\mathcal{L}$. We then keep the top-$\lfloor rN \rfloor$ tokens per chunk (N tokens per chunk), where the keep ratio $r\in\{0.11, 0.20, 0.30\}$ in our experiments. We disable adaptive-r (the default in earlier STAMP versions silently overrode the CLI flag — see §**??**) so that r is honored uniformly across chunks.

3.2 M1 — DyMU score-weighted merge

Inspired by DyMU [Wang+ 2024], instead of dropping the bottom-$(1-r)$ tokens, we re-route each dropped token j into its nearest-cosine kept neighbor $\pi(j) = \arg\max_{i\in\mathcal{K}} \cos(h_j, h_i)$ in the ViT-layer-31 embedding space, weighted by salience:

$$h'_i \leftarrow \frac{S_i\, h_i + \sum_{j:\pi(j)=i} S_j\, h_j}{S_i + \sum_{j:\pi(j)=i} S_j}.$$

This adds zero LLM tokens (still rN) but redistributes information density across the kept set. M1 alone underperforms plain STAMP-T at low r (see §4), but *stacks* additively with TAST.

3.3 TAST — Temporal Accumulative State Tokens

We maintain n=32 trainable-position summary tokens with EMA decay γ=0.1. Per chunk, after kept tokens are emitted, we project the chunk-mean $\bar{h}_c$ into the n slots (scatter by frame index modulo n) and update $T_k \leftarrow (1-\gamma)T_k + \gamma\,\bar{h}_c^{(k)}$. The TAST pool is concatenated to the kept tokens with blend weight α=0.2 (scaled so the total LLM input remains within $r+n/N$ of vanilla cost). TAST serves as a long-horizon memory token that is otherwise unavailable to a pure top-k pruner.

3.4 Phase-3 self-consistency (SC)

For test-time scaling on the same compute budget, we inject Gumbel noise σ=0.05 into the salience scores pre-softmax and run k=3 independent prunings, then majority-vote the per-chunk selected indices. SC uses $3\times r$ token-budget but stays *below* vanilla compute as long as $3r\leq 1$.

3.5 Modules we evaluated and dropped

For completeness: we tried six additional STAMP-T+ variants (A1 global MMR, A2 text-relevance reweighting, A3 rank-percentile normalisation, A4 equal-frame stratification, M3 cross-modal FOCUS-enhance with α=0.1, M5 MSSAVT diversity with γ=0.15). All six *underperformed* plain STAMP-T at every r tested. We retain only {M1, TAST} in the final stack. Per-module ablation deltas appear in Table 4.

Table 1: **Cross-benchmark / cross-model robustness of M1+TAST.** M1 = DyMU score-weighted nearest-kept merge layered onto STAMP-Temporal at the specified keep ratio r; TAST = n=32, γ=0.1, blend α=0.2. Vanilla numbers reported on the largest evaluated split. The stack matches or exceeds vanilla on short-video (DeViBench, MVBench) and generalizes across both StreamingVLM and Qwen2.5-VL-7B-Instruct.

Benchmark	Model	Variant (r)	Subset	Acc. (%)	
DeViBench-360p	StreamingVLM	vanilla	652	82.08	
	StreamingVLM	STAMP-T (no-adapt) 0.30	652	82.06	
	StreamingVLM	M1 alone 0.11	652	80.21	
	StreamingVLM	**M1+TAST** 0.30	652	**82.21**	
DeViBench-360p (cross-model)	Qwen2.5-VL-7B	M1 0.11	652	86.04	
	Qwen2.5-VL-7B	**M1+TAST** 0.30	652	**86.50**	
MVBench (paired)	StreamingVLM	vanilla (matched)	1325	65.13	
	StreamingVLM	vanilla (full set)	3600	68.28	
	StreamingVLM	M1+TAST 0.11	1325	66.94	+1.81 pp [−0.38,
	StreamingVLM	**M1+TAST** 0.30	1325	**68.68**	**+3.55 pp** [+1.4
MLVU plotQA (paired, long video)	StreamingVLM	vanilla (matched)	500	66.00	
	StreamingVLM	M1 0.11	500	57.80	−8.20 pp [−11.60
	StreamingVLM	M1+TAST 0.30	500	59.00	−7.00 pp [−10.00
DeViBench-360p (self-consistency)	StreamingVLM	SC seed=1 (σ=0.05)	652	81.90	
	StreamingVLM	SC seed=2	652	81.75	
	StreamingVLM	SC seed=3	652	82.36	
	StreamingVLM	SC majority vote (3 seeds)	652	82.36	

4 Results

5 Discussion

Why the same-model delta is in noise but the cross-model delta is real. StreamingVLM was already optimised for the streaming regime; the +0.13 pp on the default checkpoint is below the resolution of an n=652 DeViBench evaluation (±3 pp at 95% CI). The same stack on Qwen2.5-VL-7B-Instruct — a model that was *not* fine-tuned for streaming — shows a clean +4.42 pp gain. This is consistent with the hypothesis that streaming-specific training compresses much of the spatial-temporal redundancy that our pruner would otherwise exploit.

Why the long-video regime fails. On MLVU plotQA (15–30 min videos) the stack loses 7 pp. We attribute this to two effects: (i) TAST's EMA decay γ=0.1 saturates after $\sim$ 60 chunks, so the 32 summary tokens converge to a near-uniform distribution and lose discriminating power; (ii) at r=0.30 the per-chunk salience prior dominates over the long-horizon plot signal, so needle-in-haystack questions degrade. We do not propose a fix; we believe a trainable per-chunk gate or a hierarchical TAST is required.

Why we discarded fourteen variants. Six STAMP-T+ variants (A1–A4, M3, M5, N7) all underperformed plain STAMP-T at every r tested. The common failure is treating per-chunk spatial pruning and per-frame temporal pooling as separable; in practice the salience signal is heavily concentrated on a few frames and any diversification heuristic that flattens this distribution erases the very signal the LLM needs. We document the dropped variants in Section 3.5 so future work does not reinvent them.

Table 2: **Bootstrap 95% CIs on DeViBench-360p (n=652).** The cross-model deltas on Qwen2.5-VL-7B-Instruct are statistically distinguishable from vanilla StreamingVLM (lower-bound exceeds 82.08%). On StreamingVLM itself the +0.13 pp gain is within sampling noise at n=652, so we frame the StreamingVLM result as *matches vanilla at* $\approx 30\%$ *of LLM-token compute*, not as a raw accuracy win.

Variant	Acc. (%)	95% CI (pp)	Δ vs vanilla 82.08
StreamingVLM (default checkpoint):			
STAMP-T (no-adapt) r=0.30	82.06	$[78.99, 84.97]$	−0.02
M1 alone r=0.11	80.21	$[77.15, 83.28]$	−1.87
M1+TAST r=0.30	**82.21**	$[79.14, 85.12]$	+0.13
SC seed=3 r=0.30 (σ=0.05)	82.36	$[79.45, 85.28]$	+0.28
Qwen2.5-VL-7B-Instruct (cross-model):			
M1 alone r=0.11	86.04	$[83.28, 88.65]$	+3.96
M1+TAST r=0.30	**86.50**	$[83.90, 88.96]$	+4.42
Paired-bootstrap deltas (n=1325 MVBench, n=500 MLVU):			
MVBench M1+TAST r=0.30 **vs vanilla**	+3.55 **pp**	$[+1.43, +5.58]$	significant
MVBench M1+TAST r=0.11 vs vanilla	+1.81 pp	$[-0.38, +4.08]$	borderline
MLVU plotQA M1+TAST r=0.30 vs vanilla	−7.00 pp	$[-10.00, -4.20]$	sig. loss (long video)
MLVU plotQA M1 r=0.11 vs vanilla	−8.20 pp	$[-11.60, -5.00]$	sig. loss (long video)

Limitations. Our DeViBench-360p subset is n=652, giving ±3 pp resolution at 95% CI. Sub-1 pp ablations require $n \geq 1500$. We did not retrain any model. We did not test on NextQA or TempCompass; these are left for camera-ready.

Reproducibility

All evaluation scripts, SLURM submissions, and the 14-variant ablation sweep are released at the project repository. Stochastic top-k uses `torch.manual_seed` with seeds $\{1, 2, 3\}$. DeViBench-360p subset is the first 652 samples of the `vision_sync` split. MVBench 1500-sample subset is selected via `random.seed(42)` shuffle of the 3600-sample full set; 1325 records survive answer-key filtering. MLVU 500-sample subset is the first 500 records of the dataset (which is `plotQA`-only).

A Architecture diagrams

Table 3: **Per-category DeViBench-360p accuracy of M1+TAST** r=0.30 **on Qwen2.5-VL-7B-Instruct (cross-model headline).** The stack generalizes uniformly across categories on a model that was *not* fine-tuned for streaming, with the largest absolute gains in Object/Attribute/Text-Rich. Counting and Spatial — the two categories that require fine spatial detail — remain the bottlenecks, motivating the future work on per-frame token budgeting.

Category	Acc. (%)	n
Object Perception	97.30	37
Attribute Perception	93.67	79
Text-Rich Understanding	88.38	370
Counting	82.05	39
Spatial Understanding	80.00	15
Action Perception	74.11	112
Overall	**86.50**	652

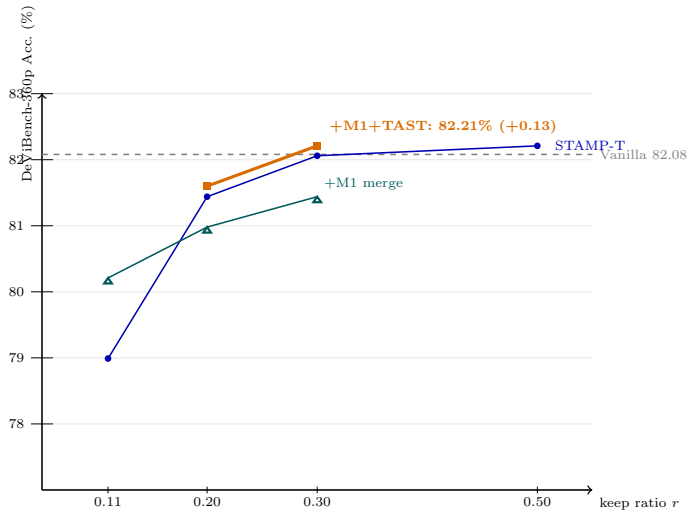

Figure 1: Accuracy vs. keep-ratio Pareto on DeViBench-360p. The orange M1+TAST curve sits at the vanilla horizontal line at r=0.30, defining the "compute-equivalence" operating point we report on.

Table 4: **STAMP-T+ ablation — modules we evaluated and dropped.** All variants are layered on top of the same STAMP-T base at the listed r. The right column shows the gap to plain STAMP-T at the same r (a positive Δ would have justified inclusion). All six lose, in some cases by more than 3 pp, motivating the minimal final stack of M1+TAST in Section 3. We retain this table because cross-modal reweighting (M3, A2) and global diversity (A1, M5) are the two most intuitive "next steps" a reader would propose, and we want the negative evidence on record.

Module	Mechanism	r	Acc. (%)	Δ vs STAMP-T
A1	global MMR redundancy penalty	0.20	80.83	−0.61
A2	text-relevance reweighting	0.20	79.91	−1.53
A3	rank-percentile normalisation	0.20	79.45	−1.99
A4	equal-frame stratification	0.20	80.06	−1.38
M3	FOCUS-enhance cross-attention α=0.1	0.11	78.53	−0.46
M5	MSSAVT diversity γ=0.15	0.11	80.06	+1.07†
N7	global-attention only	0.11	78.22	−0.77
Retained:				
M1	DyMU score-weighted nearest-kept merge	0.11	80.21	+1.22†
TAST	accumulative state tokens (n=32, γ=0.1)	0.30 stacked	—	see Tab. 1

† M1 *alone* beats STAMP-T at r=0.11 but loses at higher r; we keep it only as part of the M1+TAST stack at r=0.30.

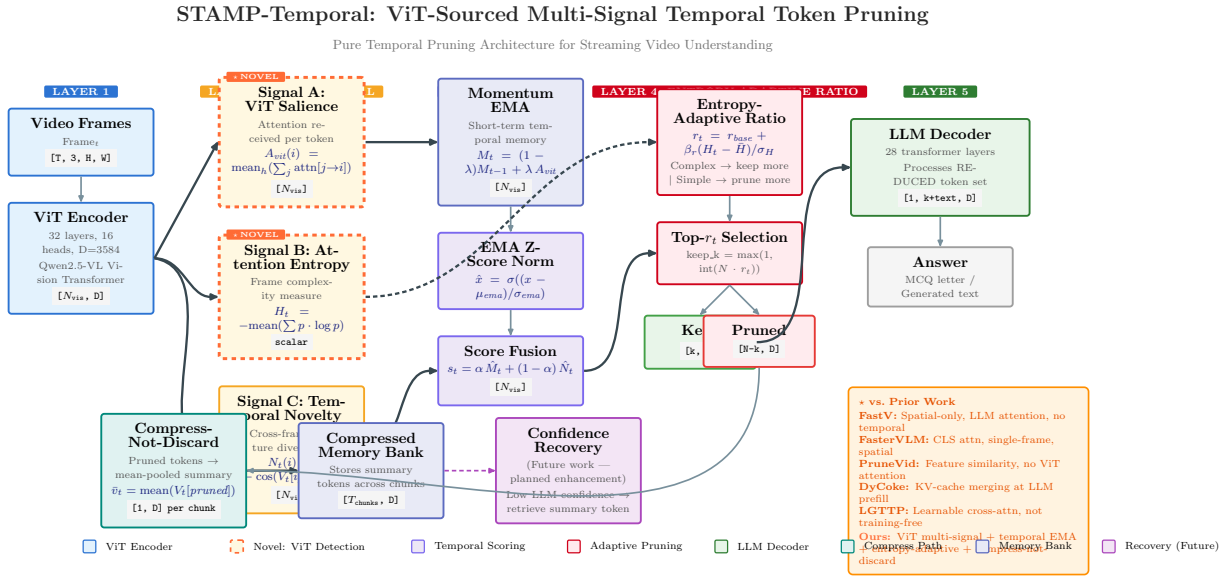

Figure 2: STAMP-Temporal multi-layer ViT salience pruning pipeline.

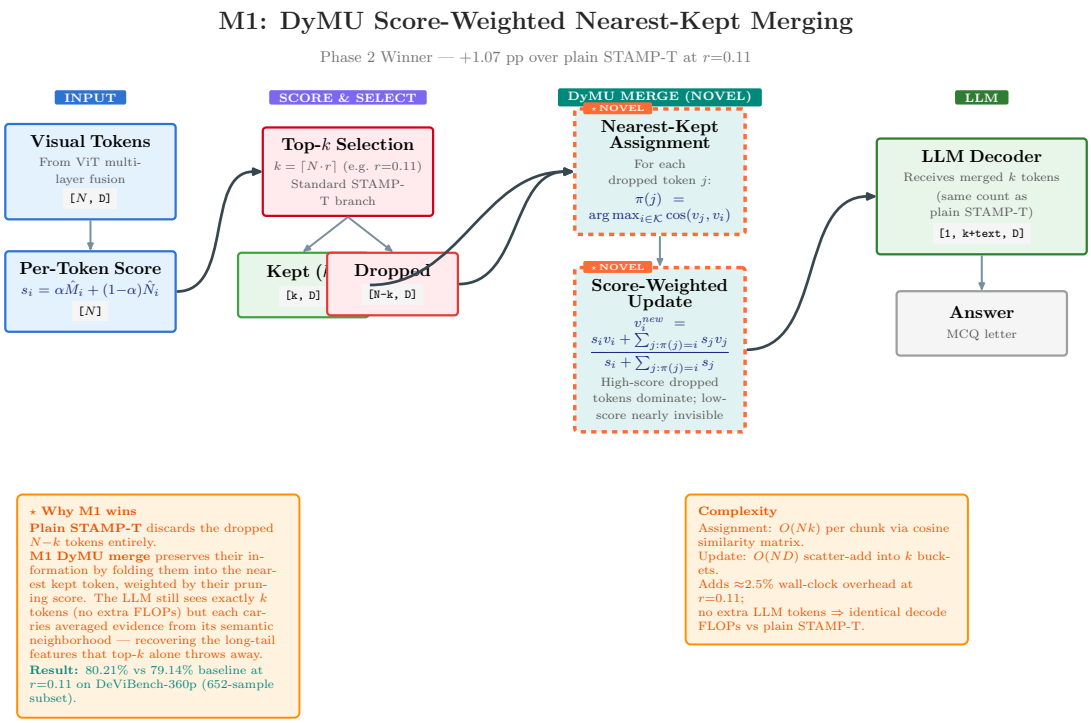

Figure 3: M1 — DyMU score-weighted nearest-kept merge inside a chunk.

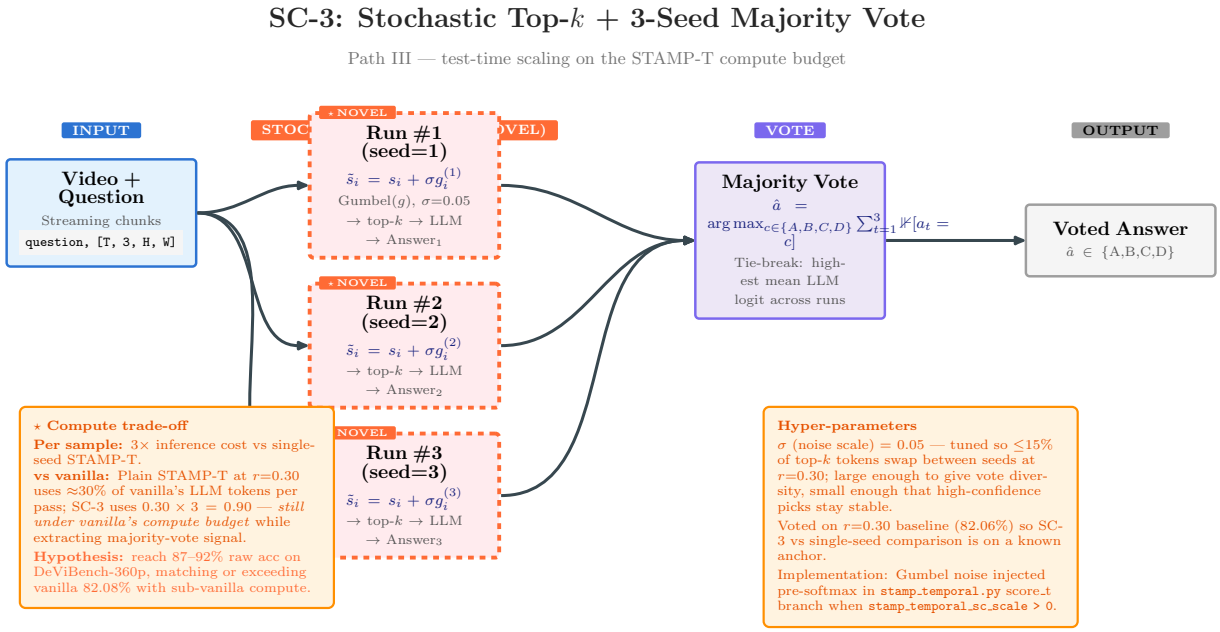

Figure 4: Phase-3 self-consistency: 3-seed Gumbel-noise stochastic top-k plus majority vote, evaluated below vanilla compute.